# What can machine-learning models trained on published hydrothermal-conversion data tell a new laboratory? A pre-registered audit with locked test studies

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## Highlights

- Pre-registered audit of literature-trained HTL/HTC models; 38 studies held out
- Yields of unseen studies were barely predictable; random splits hid this
- Laboratories agree on the direction of temperature effects, not on yield levels
- Trend-, ranking- and metadata-based shortcuts failed on held-out studies
- Study-calibrated intervals and a few local runs gave honest predictions

## Abstract

Machine-learning surrogates of hydrothermal liquefaction (HTL) and carbonization (HTC) are trained on runs compiled from the literature and usually validated by random splits. This work audited what such models can deliver for a laboratory whose study they have not seen, using a curated database of 3,693 runs from 188 studies, a pre-registered protocol and 38 studies locked away before modelling. Across 12 learners, the best bio-oil-yield RMSE rose from 7.6 wt% under random splitting to 15.7 wt% for unseen studies ($R^2$ 0.82 → 0.24), and on locked studies no learner predicted bio-oil yields substantially better than the training mean.

Study-level offsets made up 0.49–0.65 of out-of-study error variance and were not explained by recorded work-up or yield-basis descriptors. Model-free within-study comparisons showed that laboratories agree on the direction of the temperature response below 300 °C (bio-oil +0.87 wt% per 10 °C at 250–300 °C, same sign in 81% of studies), yet models predicted within-study changes no better than absolute yields, and an exploratory advantage in choosing the best planned run was not confirmed on locked studies. What held was honest uncertainty and local anchoring: conformal intervals calibrated across studies covered 0.82–0.91 of new-study runs (row-calibrated: 0.46–0.61), and three measured runs lowered the median per-study bio-oil error from 12.37 to 8.47 wt%. Literature-trained hydrothermal models should be validated by study, reported with study-calibrated intervals and anchored with local runs.

**Keywords:** hydrothermal liquefaction; hydrothermal carbonization; machine learning; conformal prediction; pre-registration; transferability

# 1. Introduction

Hydrothermal liquefaction (HTL) and hydrothermal carbonization (HTC) convert wet lignocellulosic and residual biomass into bio-oil (biocrude), a carbon-rich solid (hydrochar or char), an aqueous phase and gas, without prior drying (Toor et al., 2011; Elliott et al., 2015; Funke and Ziegler, 2010; Libra et al., 2011). Product yields depend non-linearly on feedstock composition, temperature, residence time, solid loading, catalysts and co-solvents (Gollakota et al., 2018), and experiments are costly. Machine-learning (ML) surrogates trained on data compiled from the literature have therefore become a common tool for predicting yields and product properties and for screening operating windows (Zhang et al., 2021; Leng et al., 2022; Katongtung et al., 2024; Gungor et al., 2025; Tian et al., 2025; Djandja et al., 2027). In a systematic PubMed search of ML studies on hydrothermal conversion (2015–2026; 44

primary studies, 20 of them in *Bioresource Technology*), the median of the best coefficient of determination reported in the abstracts was 0.93 (Supplementary Table S1).

Most of these surrogates are evaluated by randomly partitioning experimental runs into training and test sets. Literature-compiled data, however, are clustered: each source publication contributes a series of runs that share the feedstock batch, reactor, analytical work-up and the authors' definition of a "yield". When runs from the same study fall on both sides of a random split, the test set no longer represents a new study, and accuracy is overstated — a form of group leakage that has been documented across ML-based science (Kapoor and Narayanan, 2023) and, recently, for biochar studies (Jang et al., 2026). For HTL specifically, Elfetni et al. (2027) curated a database of 3,693 hydrothermal experiments (Elfetni, 2026), built physically admissible four-phase surrogates, and showed that publication-aware validation exposes large inter-study heterogeneity that a few local calibration runs can partly remove. Prediction intervals have also been added to HTL surrogates (Elfetni et al., 2026; Djandja et al., 2027), but their coverage has been assessed on randomly held-out runs.

Three questions that matter for anyone applying a published surrogate to a new laboratory, feedstock or plant remain open. First, how much of the error on an unseen study is a study-level offset rather than run-to-run scatter, and can it be traced to recorded differences in how studies work up and report their products (extraction solvent, yield basis, fraction definitions, reactor scale)? Bio-oil yield is an operationally defined quantity: it depends on the solvent used to recover the oil and on the basis of the reported mass fraction (Valdez et al., 2011; Watson et al., 2019). Second, do conformal prediction intervals (Vovk et al., 2022; Lei et al., 2018; Angelopoulos and Bates, 2023), whose validity rests on exchangeability between calibration and test points, keep their nominal coverage when the test point comes from a new study? Exchangeability holds across studies, not across runs of a clustered data set (Dunn et

al., 2022; Barber et al., 2023). Third, when a laboratory has measured only a handful of runs, how should these runs be combined with a literature-trained model so that both the prediction and its interval remain trustworthy?

A fourth question follows from the structure of the problem. If most of the error for a new study is a study-wide offset, *differences* between runs of the same study — how much the yield changes when temperature or residence time is changed — might transfer even when absolute yields do not. Learners that are invariant to study offsets, such as models trained on within-study deviations or pairwise-difference regression (Tynes et al., 2021), could then be useful for planning experiments, and multi-study learning has been proposed for exactly such replicability problems (Patil and Parmigiani, 2018). Comparable limitations of literature-derived data have been reported for reaction-yield prediction in chemistry (Li and Gong, 2024), but whether literature-trained models carry *trends* to new studies has, to our knowledge, not been tested for bioresource conversion.

This work audits what literature-trained models can deliver for a new laboratory, using the largest public curated hydrothermal database (Elfetni, 2026), a pre-registered analysis plan and 38 source studies locked away before any model was fitted. Five capabilities were tested, each against held-out studies: (i) predicting absolute yields; (ii) explaining study offsets from reported work-up and yield-basis metadata; (iii) predicting within-study trends, with absolute, within-study and pairwise-difference learners and with a model-free analysis of runs that differ only in temperature or residence time; (iv) choosing the best of a study's planned runs, compared with rules of thumb; and (v) providing honest uncertainty and exploiting a few local runs. For (v) the study assembles established statistical constructions — study-weighted conformal calibration for hierarchical data (Dunn et al., 2022), empirical-Bayes shrinkage of a study offset (Efron and Morris, 1975) and re-calibration over exchangeable calibration groups as in meta-learned and small-area conformal prediction (Fisch et al., 2021; Bersson and Hoff,

2024) — into a workflow (STRATA). All pre-registered hypotheses are reported, including those that failed. The evidence is condensed into a capability matrix, a model-free trend atlas and a minimum validation checklist for literature-trained bioresource surrogates. Physical admissibility of multi-phase predictions, already addressed by Elfetni et al. (2027), is not a contribution of this study.

# 2. Materials and methods

## 2.1. Data

The primary data were the Version 1.0.0 release of the curated hydrothermal treatment/liquefaction database (Elfetni, 2026) (CC BY 4.0; 3,693 runs from 188 source publications, 1982–2026; SHA-256 of the file verified at every load). Each run records the source DOI, feedstock ultimate, proximate and lignocellulosic composition, operating conditions, catalyst, reaction medium, the reported yield basis and separation procedure, and product yields. Two targets were modelled: bio-oil yield (HTL and solvothermal HTL runs) and solid-product yield (HTL, solvothermal HTL and HTC runs), both in wt%. Pre-specified exclusions were: aqueous-phase reforming runs; yields outside 0–100 wt%; yields reported on a carbon basis; missing feedstock C or O, temperature or residence time; and studies contributing a single run to a target (Table 1). An independent external data set — a 4TU.ResearchData hydrochar database (Abdeldayem et al., 2025) (CC BY 4.0) without study identifiers — was used only for external validation of solid yield (Section 2.6).

**Table 1.** Data used for each target after pre-specified exclusions.

| Target | Split | Runs | Studies | Median runs/study | Yield mean ± SD (wt%) | Excluded (carbon basis / missing inputs / single-run study) |
|---|---|---|---|---|---|---|
| Bio-oil | Development | 2261 | 122 | 15 | 30.8 ± 18.0 | 3 / 11 / 4 |

| Target | Split | Runs | Studies | Median runs/study | Yield mean ± SD (wt%) | Excluded (carbon basis / missing inputs / single-run study) |
|---|---|---|---|---|---|---|
| Bio-oil | Locked test | 556 | 28 | 12 | 26.9 ± 13.1 | 0 / 1 / 0 |
| Solid | Development | 2156 | 120 | 13 | 33.4 ± 20.2 | 0 / 12 / 5 |
| Solid | Locked test | 457 | 30 | 12 | 40.0 ± 22.1 | 0 / 0 / 1 |

## 2.2. Locked study-level test set and leakage controls

Before any model was fitted, 20% of the source publications (38 studies, seed 20261110) were drawn at random and locked (`locked_test_studies.json`, committed to version control before the first model fit). They were evaluated once, after all modelling decisions had been frozen, with models trained on all development studies. The analysis protocol, research questions, metrics, statistical tests and seeds were registered in the repository before experiments were run; three deviations are documented in Supplementary Section S2. Leakage controls were: the unit of splitting is the source publication; DOI, title and publication year are never model inputs (year is used only to define the temporal split); all statistics-estimating preprocessing (median imputation with missing-value indicators, standardisation) is refitted inside each training fold; the external database was screened for runs that duplicate runs of the primary database (36 potential duplicates removed); and the primary database was audited for identical runs reported in more than one source study (Section 2.6, E10).

## 2.3. Features

*Base features* describe chemistry and conditions: feedstock C, H, O, N and ash; lignin, cellulose and hemicellulose; temperature; log residence time; solid loading; heating rate; catalyst class (alkali carbonate, alkali hydroxide, acid, metal/supported metal, other) and catalyst-to-biomass ratio; reaction-medium class (water, alcohol, water–alcohol, water with hydrogen donor or co-reactant, ketone, other organic); gas pre-pressurisation; and process

subtype. *Protocol/reporting descriptors* (P) encode how a study defined and recovered its products, parsed deterministically from the curated yield-basis and separation fields: dry-ash-free basis; basis including co-reactant or solvent mass; product defined as several combined fractions; aqueous or gas phase obtained by difference; extraction solvent (dichloromethane, acetone, ethyl acetate, diethyl ether, tetrahydrofuran, alcohol, hydrocarbon, other chlorinated, none named); separation route not stated; and log reactor volume. Parsing rules are in `src/data/load.py` and unit-tested.

## 2.4. Learners, tuning and evaluation schemes

Twelve learners were compared: mean predictor, ridge regression, *k*-nearest neighbours, support-vector regression, multilayer perceptron, random forest (Breiman, 2001), extra trees (Geurts et al., 2006), histogram gradient boosting (Pedregosa et al., 2011), LightGBM (Ke et al., 2017), XGBoost (Chen and Guestrin, 2016), CatBoost (Prokhorenkova et al., 2018) and GPBoost, which adds a study-level random intercept to tree boosting (Sigrist, 2022). A tabular foundation model (Hollmann et al., 2025) was pre-specified but could not be run because its weights require a personal licence acceptance (Supplementary Section S2). Every learner received the same tuning budget — 20 random hyper-parameter configurations scored by study-grouped five-fold cross-validation (CV) on the development studies — separately for each feature set; the selected configuration was then frozen. Each learner was evaluated by (a) random-row five-fold CV, the prevailing practice, and (b) study-grouped ten-fold CV repeated with three pre-specified seeds, in which every development study is predicted by a model that has never seen any of its runs (Roberts et al., 2017). Metrics were RMSE, MAE and $R^2$ over all runs and the RMSE of each study.

## 2.5. STRATA

STRATA wraps any point predictor $f$. Let $r_{ij} = y_{ij} - f(x_{ij})$ be the residual of run $i$ of study $j$ obtained *out of study*, i.e. from a model that did not see study $j$ (inner study-grouped five-fold CV on the training studies).

*Heterogeneity decomposition.* Residuals are modelled as $r_{ij} = u_j + \varepsilon_{ij}$ with study offset $u_j$ (variance $\tau^2$) and within-study error $\varepsilon_{ij}$ (variance $\sigma_w^2$). Both are estimated by the unbalanced one-way ANOVA method of moments (DerSimonian and Laird, 1986),

$$\hat{\sigma}_w^2 = MS_W,\ \hat{\tau}^2 = max\left(0, \frac{MS_B - MS_W}{n_0}\right),\ n_0 = \frac{N - \sum_j n_j^2 / N}{J - 1},$$

and the between-study share is $\hat{\tau}^2 / (\hat{\tau}^2 + \hat{\sigma}_w^2)$, analogous to $I^2$ in meta-analysis (Higgins and Thompson, 2002); 95% CIs are obtained by resampling studies (2,000 cluster-bootstrap replicates (Efron, 1979)). To test whether P explains study offsets, the mean out-of-study residual of each development study (base-feature model) was regressed on the study means of P by ridge regression with leave-one-study-out prediction; significance was assessed by permuting offsets across studies (1,000 permutations).

*Study-calibrated conformal intervals (SC).* For a new run the interval is $f(x) \pm \hat{q}$, where $\hat{q}$ is the weighted $(1 - \alpha)$ empirical quantile of $|r_{ij}|$ in which every calibration study receives the same total weight (run weight $1/n_j$), with the usual finite-sample correction. This is the CDF-pooling construction for two-layer hierarchical data of Dunn et al. (2022), under which the target of validity is a run from a new study that is exchangeable with the calibration studies; its guarantee is approximate rather than exact in finite samples, and it holds marginally over studies, not for every individual study. As in cross-validation-based conformal methods, the calibration residuals come from inner models fitted on subsets of the training studies, whereas

test predictions come from the model fitted on all training studies — a standard, slightly conservative approximation. The comparator, *row-calibrated* intervals (RC), uses the unweighted quantile of residuals from random-row CV — the construction implied when conformal calibration is performed on a random split.

*k-shot empirical-Bayes adaptation (KS).* When $k$ runs of the new study have been measured, the predictions for its remaining runs are shifted by

$$\hat{\delta}_k = \frac{k}{k+\hat{\lambda}} \acute{r}_S, \hat{\lambda} = \hat{\sigma}_w^2 / \hat{\tau}^2,$$

where $\acute{r}_S$ is the mean residual of the $k$ support runs. This is the posterior mean of $u_j$ under a normal random-intercept model, a James–Stein-type shrinkage (Efron and Morris, 1975) that tends to the raw mean offset as $k$ grows and to zero when studies are homogeneous. The interval half-width $\hat{q}_k$ is re-calibrated by applying exactly this procedure to the calibration studies (20 random support sets per study) and taking the study-weighted quantile of the adjusted residuals of their query runs. The ablation compares no offset ($k=0$), the raw mean offset and the shrunk offset.

## 2.6. Experiments

E1/E2 (accuracy and baselines) applied the two evaluation schemes to all learners and both feature sets. E3 (STRATA) used the study-grouped outer folds: for each fold, inner random-row and inner study-grouped CV on the training studies produced RC and SC calibration residuals and $\hat{\lambda}$; the outer test studies were then predicted, and for studies with at least 11 runs the *k*-shot procedure was evaluated for $k \in \{1,2,3,5,10\}$ with 20 random support sets per study (identical studies for all $k$). The base learner for STRATA was fixed by a pre-specified rule as the native-missing-value learner with the lowest study-grouped CV RMSE on bio-oil yield (XGBoost). Robustness (E4) was assessed by adding Gaussian noise (5–20% of the

training standard deviation) to continuous test features and by training on 25–100% of the training studies. Generalisation (E5) was assessed on the locked test studies, temporally (studies published up to 2018 → studies published after 2018), by leaving out one feedstock family at a time (families with ≥ 3 studies), and externally on the 4TU hydrochar database (solid yield; runs sharing an identical feedstock composition treated as one pseudo-study). E6 measured single-thread fit and prediction time, peak resident memory and serialized model size. E7 varied α (0.05, 0.10, 0.20), the $\tau^2$ estimator (method of moments vs. restricted maximum likelihood), the number of outer folds (5 vs. 10) and tuning (tuned vs. library defaults). E8 analysed out-of-study errors by feedstock family, process, extraction solvent, temperature, catalyst use and yield basis. Three further analyses were added during internal review, after the main results were known, and are reported as such: E9 repeated the STRATA evaluation with six other base learners, including the training mean as a 'no-ML' reference (development folds only); E10 audited runs that occur with identical feedstock composition, conditions and yield in more than one source study, and repeated the study-grouped evaluation without them; E11 measured the wall-clock overhead of STRATA.

## 2.7. Trend, ranking and anchoring analyses (pre-registered addenda)

These analyses were registered as protocol addenda after the main analyses had been completed and before they were run (Supplementary Section S2): Addendum A (E12) before any trend analysis, Addendum A2 before analysing two product-quality targets, and Addendum C before any locked-study evaluation of E12 or E14. Addendum B (E14) was registered after the development results of E12 had been seen and is exploratory; its main finding was then tested on the locked studies under Addendum C. The model-free analysis E13 was not registered and is descriptive. The commit history establishing this order is given in Supplementary Table S16.

*Within-study contrasts (E12).* For two runs *a* and *b* of the same study, the contrast is $\Delta_{ab} = y_a - y_b$; a study offset cancels in $\Delta_{ab}$. Three learners predicted contrasts for every pair of runs within each unseen study: the absolute model A, by differencing, $\hat{\Delta}_{ab} = f(x_a) - f(x_b)$; a within-study model W, trained on study-centred targets $y - \acute{y}_{\text{study}}$ with features $[x, x - \acute{x}_{\text{study}})$ (the study means of the inputs are known for a new study before any run is measured); and a pairwise-difference model PD (Tynes et al., 2021), trained on pairs of runs of the same training study with inputs $[x_a, x_b, x_a - x_b)$ and target $\Delta_{ab}$ (at most 30 random partners per run), antisymmetrised as $\hat{\Delta}_{ab} = 1/2[h(a,b) - h(b,a))$. All three used the E1 hyper-parameters of XGBoost, extra trees and LightGBM. Contrast accuracy was summarised by $R^2_{\Delta} = 1 - \sum(\hat{\Delta} - \Delta)^2 / \sum \Delta^2$, i.e. relative to the 'no-change' predictor. Hypotheses: contrasts are predicted better than levels (T1); W and PD predict contrasts better than A (T2, one-sided Wilcoxon on per-study contrast RMSE, Holm); anchoring on *k* = 1–3 measured runs, $\hat{y}_i = 1/k \sum_a (y_a + \hat{\Delta}_{ia})$, beats STRATA's shrunk offset (T3); study-calibrated conformal intervals for contrasts are valid and narrower than for levels (T4); and models rank the runs of a study usefully (T5). The same analysis was repeated for bio-oil higher heating value (HHV; values computed by correlation excluded; 1,019 development runs from 106 studies, 245 locked runs) and bio-oil carbon content (1,036 runs, 110 studies; 227 locked runs) (Addendum A2; XGBoost tuned with the E1 budget).

*Model-free trends (E13; descriptive, not pre-registered).* Within each development study, pairs of runs that differ only in temperature (all other base features and the feedstock identical) or only in residence time were identified. Their observed slopes (change in yield per +10 °C, or per doubling of residence time) were summarised per study and temperature band; medians across studies were bootstrapped over studies, and the fraction of studies whose slope has the same sign as the median was recorded.

*Contrast types and choosing conditions (E14, exploratory; confirmed under Addendum C).* Contrast accuracy was split into temperature-only, residence-time-only and other pairs. For each unseen study with ≥ 5 runs, the run with the highest predicted yield (A: highest $f$; W: highest $g$; PD: highest mean predicted contrast against the other runs) was compared with a rule of thumb — for bio-oil, the run with temperature closest to 325 °C, the centre of the optimum band in E13; for solids, the lowest-temperature run — and with random choice, using the regret $max_i\, y_i - y_{\text{chosen}}$. The bio-oil comparison was then tested once, one-sided, on the locked studies (Holm over the three learners).

## 2.8. Statistical analysis

The unit of analysis for all inference was the source study. Learners and STRATA variants were compared through per-study RMSE (averaged over seeds) with the Wilcoxon signed-rank test (Wilcoxon, 1945); families of comparisons were corrected with Holm's step-down procedure as described by Demšar (2006). Effect sizes are the median paired difference with a 95% bootstrap CI over studies and the matched-pairs rank-biserial correlation. One-sided tests were used only for directional hypotheses registered in advance (P reduces error; SC coverage exceeds RC coverage). Coverage was summarised both over runs and with equal study weights, with 95% cluster-bootstrap CIs; on development folds, point estimates average the three outer seeds and CIs were computed for the first seed. Statistical significance is claimed only for Holm-adjusted $p < 0.05$.

## 2.9. Software and reproducibility

All analyses ran on a 4-core CPU without GPU using Python 3.13, scikit-learn, LightGBM, XGBoost, CatBoost, GPBoost, statsmodels and SciPy (exact versions in `requirements.txt`). A single command (`python run_all.py --mode full`) regenerates every number, table and figure from the vendored data; a quick mode, run from a

fresh clone of the repository, verified the whole pipeline in 12 minutes on the same machine, and the full analysis stages reproduced every value in this article exactly. All values reported in this article are inserted programmatically from the generated result files.

# 3. Results and discussion

## 3.1. Random splits overstate accuracy for new studies

Fig. 1 summarises the study design and Fig. 2 the data: most studies contributed between a few and a few dozen runs, a few contributed more than 100, and the source studies span four decades. Under random-row CV the gradient-boosting learners reproduced the accuracy typically reported in the literature (best bio-oil $R^2 = 0.82$, RMSE = 7.6 wt%, LightGBM; best solid-yield RMSE = 7.0 wt%) (Table 2, Fig. 3). When every study was predicted by a model that had not seen it, RMSE roughly doubled for the flexible learners (tree ensembles, boosting, k-NN, SVR, MLP): the ratio of study-grouped to random-row RMSE ranged from 1.52 to 2.61 for bio-oil (median 2.02) and from 1.50 to 2.28 for solid yield (median 1.97). The best study-grouped RMSE was 15.7 wt% for bio-oil (extra trees, $R^2 = 0.24$) and 13.7 wt% for solid yield (LightGBM, $R^2 = 0.54$), compared with 18.1 and 20.6 wt% for a model that always predicts the training mean. Ratios close to one were obtained only by learners that fit the data poorly in both schemes (mean predictor, ridge regression, GPBoost), i.e. the optimism of random splitting grows with the capacity of a learner to memorise study-specific patterns. Parity plots show that the near-diagonal agreement under random splitting is replaced by wide, partly systematic scatter for unseen studies (Supplementary Fig. S4).

**Table 2.** Random-row versus study-grouped cross-validation on development studies (base features; grouped CV averaged over 3 seeds).

| Target | Learner | Random CV RMSE | Random CV $R^2$ | Grouped CV RMSE | Grouped CV $R^2$ | RMSE ratio (grouped/random) |
|---|---|---|---|---|---|---|
| Bio-oil | Extra trees | 8.64 | 0.77 | 15.73 | 0.237 | 1.82 |
| Bio-oil | SVR | 10.38 | 0.668 | 15.8 | 0.23 | 1.52 |
| Bio-oil | XGBoost | 7.81 | 0.812 | 15.8 | 0.23 | 2.02 |
| Bio-oil | Random forest | 8.45 | 0.78 | 16.01 | 0.21 | 1.89 |
| Bio-oil | LightGBM | 7.61 | 0.821 | 16.11 | 0.2 | 2.12 |
| Bio-oil | k-NN | 10.27 | 0.675 | 16.26 | 0.185 | 1.58 |
| Bio-oil | CatBoost | 7.68 | 0.818 | 16.54 | 0.156 | 2.15 |
| Bio-oil | HistGB | 7.8 | 0.812 | 16.74 | 0.136 | 2.15 |
| Bio-oil | Ridge | 15.02 | 0.304 | 17.51 | 0.054 | 1.17 |
| Bio-oil | GPBoost | 17.32 | 0.075 | 17.61 | 0.043 | 1.02 |
| Bio-oil | Mean | 18.02 | -0.001 | 18.12 | -0.013 | 1.01 |
| Bio-oil | MLP | 9.19 | 0.74 | 23.95 | -0.77 | 2.61 |
| Solid | LightGBM | 6.96 | 0.882 | 13.7 | 0.541 | 1.97 |
| Solid | XGBoost | 7.02 | 0.88 | 14.14 | 0.511 | 2.02 |
| Solid | HistGB | 7.03 | 0.879 | 14.15 | 0.511 | 2.01 |
| Solid | Random forest | 8.61 | 0.819 | 14.42 | 0.492 | 1.67 |
| Solid | CatBoost | 7.12 | 0.876 | 14.48 | 0.488 | 2.03 |
| Solid | k-NN | 9.84 | 0.764 | 14.77 | 0.467 | 1.5 |
| Solid | SVR | 9.56 | 0.777 | 15.01 | 0.45 | 1.57 |
| Solid | Extra trees | 8.23 | 0.835 | 15.07 | 0.445 | 1.83 |
| Solid | Ridge | 13.96 | 0.524 | 16.34 | 0.348 | 1.17 |
| Solid | MLP | 8.57 | 0.821 | 19.51 | 0.07 | 2.28 |
| Solid | GPBoost | 19.13 | 0.106 | 19.81 | 0.042 | 1.04 |
| Solid | Mean | 20.25 | -0.001 | 20.63 | -0.039 | 1.02 |

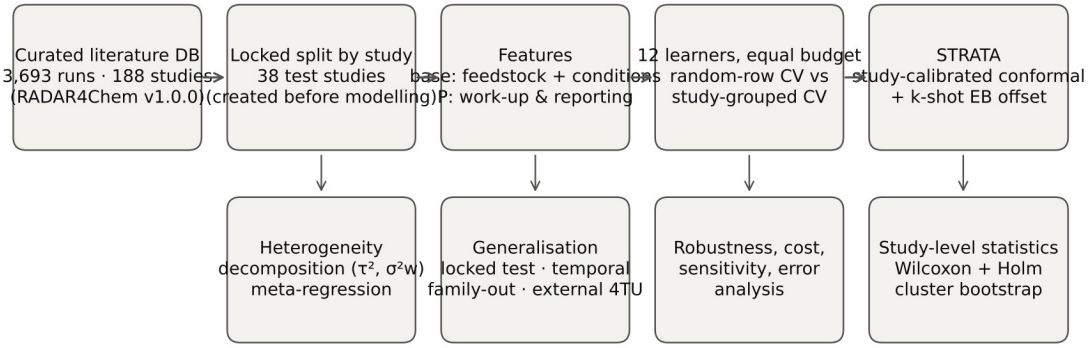



***Fig. 1.*** *Study design. A locked set of 38 source studies is held out before modelling; all development analyses use study-grouped folds; STRATA adds study-calibrated conformal intervals and k-shot empirical-Bayes adaptation to any point predictor.*


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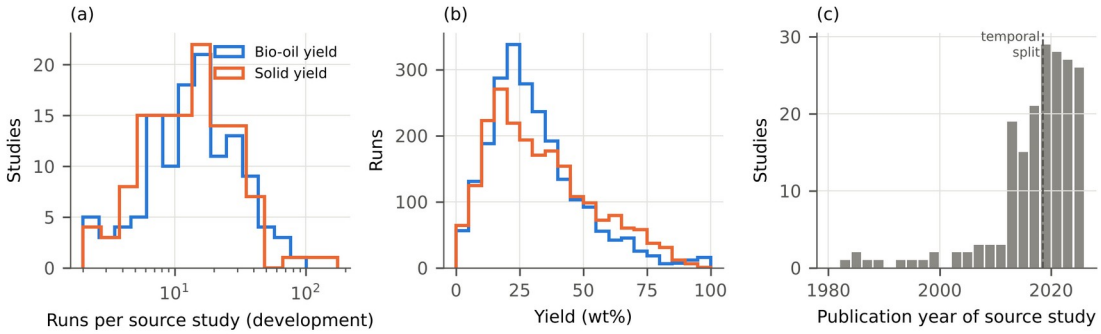



***Fig. 2.*** *Development data. (a) Number of runs contributed per source study; (b) yield distributions; (c) publication years of all 188 source studies (dashed line: temporal split).*

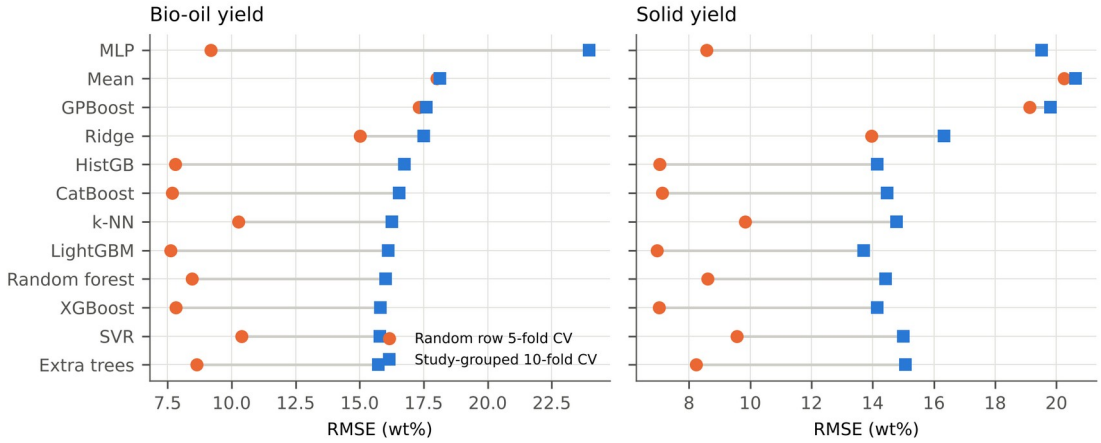



***Fig. 3.*** *Root-mean-square error of 12 learners under random-row five-fold CV (circles) and study-grouped ten-fold CV (squares; mean of three seeds), base features.*

Among learners, differences under study-grouped CV were small relative to the gap between schemes: for bio-oil, 9 learners were not significantly worse than the best one (extra trees) after Holm correction on per-study RMSE (Supplementary Table S3), and the multilayer perceptron extrapolated poorly to unseen studies (RMSE 24.0 wt%) despite competitive random-split accuracy (9.2 wt%). GPBoost, which models a study random intercept, gained nothing for unseen studies: because feedstock descriptors are nearly constant within a study, the random intercept absorbed most of the variance and its fixed-effect component — the only part available for a new study — was nearly flat (Supplementary Section S2). This illustrates


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that, in literature-compiled data, study identity and feedstock chemistry are strongly confounded, which limits how much of the between-study variation any model can attribute to chemistry.

## 3.2. Most out-of-study error is a study-level offset

For the STRATA base learner (XGBoost), study offsets were large and spread on both sides of zero (Fig. 4). The between-study share of out-of-study residual variance was 0.65 (95% CI 0.53–0.76) for bio-oil and 0.49 (0.40–0.61) for solid yield; the standard deviation of study offsets ($\hat{\tau}$) was 12.5 and 9.7 wt%, against within-study residual standard deviations ($\hat{\sigma}_w$) of 9.3 and 9.9 wt% (Table 3). A model that is well calibrated on average can therefore be biased by more than 10 wt% for an individual study, and errors for runs of the same study are strongly correlated — which is why randomly held-out runs, whose study offsets were already learnt, give an optimistic picture.

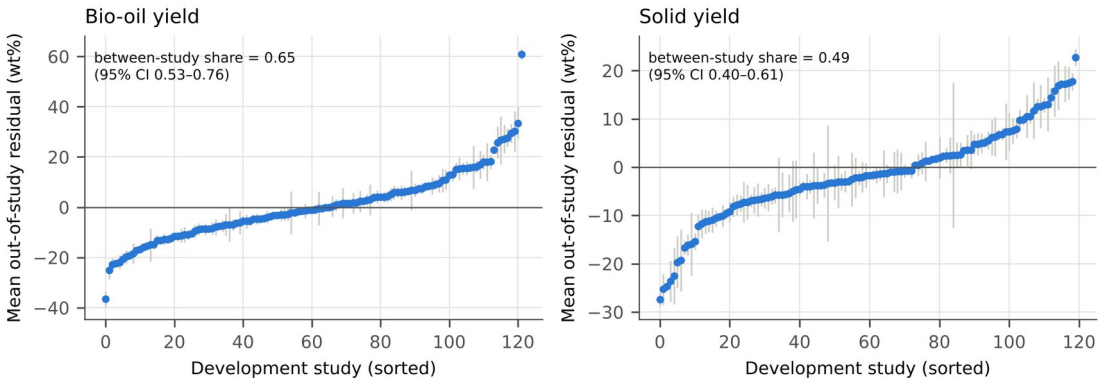



***Fig. 4.*** *Mean out-of-study residual (observed − predicted) of each development study with 95% intervals of the mean, STRATA base learner with base features. The annotated share is the between-study fraction of residual variance with its study-cluster bootstrap CI.*


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## 3.3. Work-up and reporting descriptors do not explain the offsets

Because bio-oil yield is operationally defined by the recovery solvent and the reporting basis (Valdez et al., 2011; Watson et al., 2019), study offsets were expected to depend on the recorded work-up. However, adding the parsed work-up and reporting descriptors (P) did not significantly reduce the error for unseen studies for either target after Holm correction. Adding the protocol descriptors changed the median per-study RMSE from 12.45 to 12.84 wt % for bio-oil (median paired difference 0.04 wt%, 95% CI -0.38 to 0.60; one-sided Wilcoxon $p = 0.804$; $n = 122$ studies) and from 10.86 to 11.05 wt% for solid yield ($p = 0.545$) (Table 3; all learners with P features in Supplementary Table S2). On the locked test studies, P significantly reduced per-study RMSE for 0 and significantly increased it for 0 of the 24 learner–target combinations (Holm-corrected Wilcoxon tests). Consistently, a ridge meta-regression of the study offsets on the study-level descriptors had no out-of-study predictive value (leave-one-study-out $R^2 = -0.031$, permutation $p = 0.714$ for bio-oil; $R^2 = -0.047$, $p = 0.910$ for solids), and offsets did not differ significantly between the dominant extraction solvents (Kruskal–Wallis $p = 0.104$ for bio-oil; Kruskal–Wallis $p = 0.951$ for solids). The most negative median offset for bio-oil occurred for studies using ethyl acetate (median offset -12.7 wt%, 7 studies), but such groups are small and the differences were not significant.

**Table 3.** Decomposition of out-of-study error and contribution of work-up/reporting descriptors (P).

| Target | $\tau^2$ (between) | $\sigma^2 w$ (within) | Between-study share [95% CI] | Median study RMSE base → base+P | Δ median [95% CI]; p (one-sided) | Rank-biserial | Meta-regression LOSO $R^2$ (perm. p) |
|---|---|---|---|---|---|---|---|
| Bio-oil | 156.7 | 85.6 | 0.65 [0.53, 0.76] | 12.45 → 12.84 | 0.04 [-0.38, 0.60]; 0.804 | 0.09 | -0.031 (0.714) |
| Solid | 94.6 | 97.7 | 0.49 [0.40, 0.61] | 10.86 → 11.05 | 0.05 [-0.15, 0.25]; 0.545 | 0.01 | -0.047 (0.910) |

This negative result has two readings, which the data cannot separate. Either work-up conventions matter less than chemistry and conditions that the curated fields do not capture

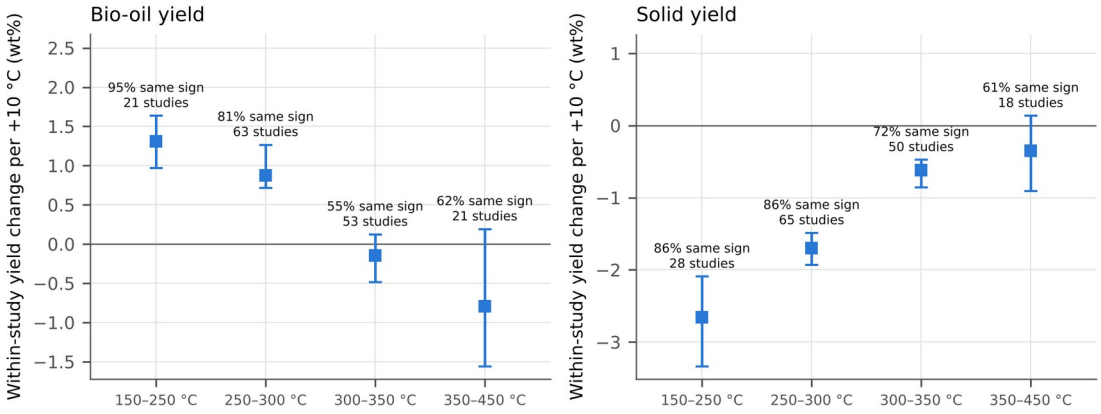

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(feedstock batch, particle size, reactor heat-up, quenching and product-handling details), or the coarse, partly missing descriptors that can be parsed from publications are too noisy to reveal their effect — 23% of development bio-oil runs named no extraction solvent. In both cases the practical conclusion is the same: the offset of a new study is not predictable from what studies currently report, so it has to be either measured (Section 3.7) or represented in the uncertainty (Section 3.6).

## 3.4. Laboratories agree on the direction of temperature effects, but models do not predict within-study changes better than levels

Runs that differed *only* in temperature occurred in 63 development studies for bio-oil and 65 for solids in the 250–300 °C band alone (2794 and 3076 pairs over all bands). Their observed slopes trace the familiar behaviour of hydrothermal conversion (Gollakota et al., 2018; Funke and Ziegler, 2010) remarkably consistently across laboratories (Fig. 5, Supplementary Table S11): bio-oil yield changed by +1.31 wt% per 10 °C below 250 °C (95% CI +0.97 to +1.64; same sign in 95% of studies) and by +0.87 wt% at 250–300 °C (81% same sign), levelled off at 300–350 °C (-0.14 wt%, -0.48 to +0.12) and tended to decrease above 350 °C (-0.79 wt%); solid yield changed by -2.66, -1.70, -0.62 and -0.35 wt% per 10 °C in the four bands (86% same sign below 300 °C). Doubling the residence time had no consistent effect on bio-oil yield (-0.06 wt%, -0.66 to +0.61; 50% same sign) and a small negative effect on solid yield (-0.49 wt%, -1.02 to -0.13). The same directions held within each process type — below 300 °C every process-specific median slope was positive for bio-oil (78–100% of studies with the same sign) and negative for solids (82–100%) (Supplementary Table S14). The direction of the temperature response is therefore shared by laboratories whose absolute yields differ by tens of wt%; its magnitude is not — the between-study standard deviation of the bio-oil slope at 250–300 °C (2.3 wt% per 10 °C) is 2.6 times the median slope.


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374

375 ***Fig. 5.*** *Model-free trend atlas. Median change in yield per +10 °C across development studies,*
376 *computed from pairs of runs of the same study that differ only in temperature (all other inputs and the*
377 *feedstock identical); error bars are 95% study-bootstrap CIs, annotations give the share of studies*
378 *whose slope has the same sign as the median.*

379 Contrary to the pre-registered hypotheses, the learners did not exploit this regularity better
380 than they predicted absolute yields (Fig. 6, Supplementary Table S12). For unseen
381 development studies, contrast $R^2_\Delta$ was 0.14–0.22 for bio-oil (level $R^2$ of the same studies 0.15–
382 0.19) and 0.25–0.43 for solids (levels 0.36–0.47) across the three learners and three model
383 types (T1 not supported). Pooled contrast $R^2$ weights large studies by their number of pairs;
384 weighting studies equally, the median per-study $R^2_\Delta$ was only 0.06–0.10 for bio-oil and 0.23–
385 0.32 for solids, and -0.28 to -0.04 for bio-oil on the locked studies (Supplementary Table
386 S15). Offset-invariant learners were better than differencing the absolute model in 1 of 12
387 Holm-corrected comparisons (T2 not supported). Even for temperature-only pairs, $R^2_\Delta$ was
388 0.14–0.18 for bio-oil and 0.48–0.52 for solids (XGBoost, models A and PD), and residence-
389 time-only contrasts were not predicted at all ($R^2_\Delta$ = 0.01 for bio-oil, model A). On the locked
390 studies the pattern was mixed: offset-invariant models retained contrast skill where the
391 absolute model failed (bio-oil $R^2_\Delta$ -0.30 for A vs 0.21 for W and 0.30 for PD; temperature-only
392 pairs 0.15 vs 0.59), but two studies accounted for 65% of the squared contrast error of A, and

per-study comparisons were not significant (0 of 12 after Holm; smallest unadjusted $p$ = 0.025). For the two product-quality targets the results were equally mixed: bio-oil HHV contrasts were predicted better than levels in development ($R^2_\Delta$ 0.39 vs level $R^2$ 0.21) but not on locked studies (-0.10 vs 0.25); for bio-oil carbon content, $R^2_\Delta$ was 0.11–0.17 (development) and -0.07–0.22 (locked).

The two analyses are reconciled by the size of the within-study variation: because laboratories agree on direction but not on magnitude, and most within-study pairs also differ in catalyst, medium or feedstock rather than temperature alone, the within-study residual variance ($\hat{\sigma}_w$ = 9.3 wt% for bio-oil) is large compared with the systematic within-study differences that the inputs explain. Directional knowledge of the kind shown in Fig. 5 is therefore a robust prior for experimental design, but literature-trained models should not be trusted to quantify the effect of a planned change in a new laboratory.

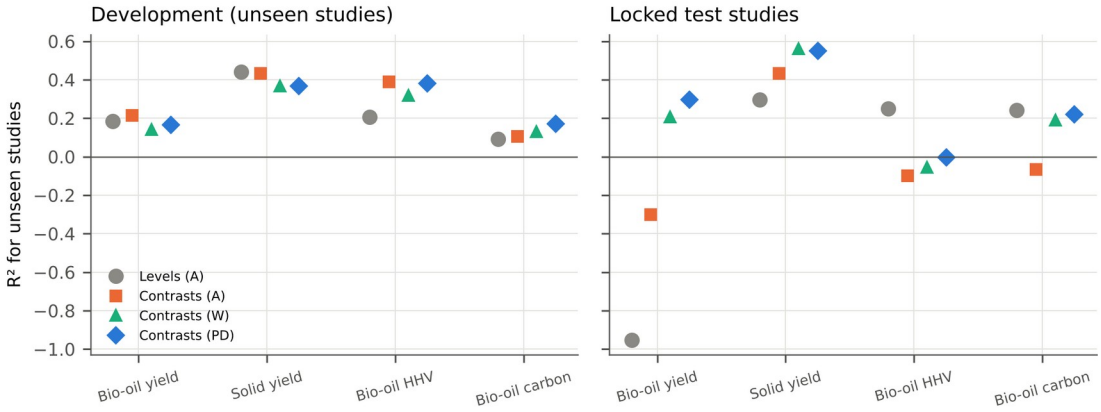



***Fig. 6.*** *Accuracy of absolute-yield predictions (levels, model A) and of within-study contrasts (models A, W and PD; XGBoost) for unseen development studies and for the locked test studies, for four targets. $R^2$ of contrasts is relative to the no-change predictor.*


20

## 3.5. Choosing the best planned run: an exploratory advantage that did not replicate

Ranking the runs of a study is a weaker requirement than predicting them. For unseen development studies, all three model types picked bio-oil conditions with lower regret than the 325 °C rule of thumb (median regret 7.0–8.7 vs 12.8 wt%; random choice 12.7 wt%; 110 studies; Holm-corrected $p < 0.001$ for model A), whereas for solids none beat the lowest-temperature rule (median regret 1.3–3.0 vs 2.2 wt%) (Fig. 7, Supplementary Table S13). Because this comparison was chosen after the E12 results had been seen, it was registered as a confirmatory test on the locked studies before they were analysed. The advantage was not confirmed: on 25 locked bio-oil studies the median regret was 11.9–12.5 wt% for the models, 14.6 wt% for the rule and 12.5 wt% for random choice (one-sided Holm-corrected $p = 0.661$ for PD). Literature-trained models should therefore not be relied on to select the best conditions of a new HTL campaign without local confirmation.

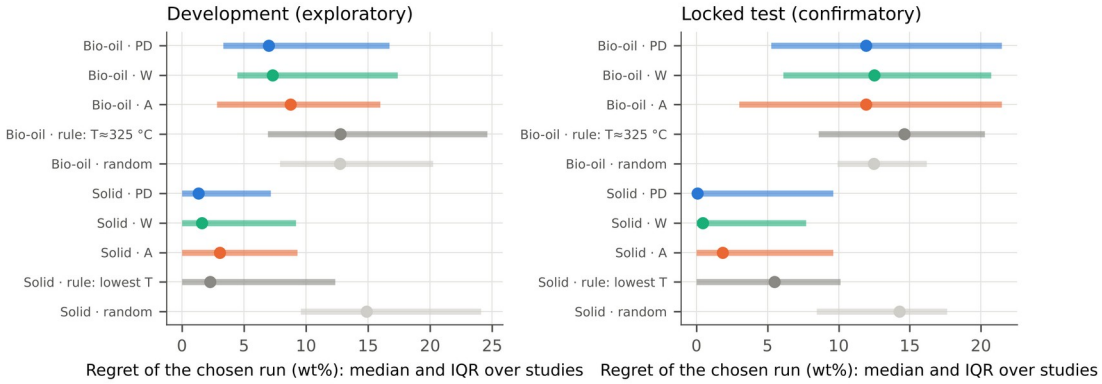



***Fig. 7.*** *Regret of choosing the run predicted to give the highest yield within a study (observed best minus observed yield of the chosen run; median and interquartile range over studies) for models A, W and PD, a rule of thumb and random choice. Left: development studies (exploratory); right: locked studies (pre-registered confirmatory test).*


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## 3.6. Row-calibrated intervals under-cover new studies; study calibration largely restores coverage

Intervals calibrated on randomly held-out runs are narrow because the residuals of seen studies omit the study offset. For unseen development studies, nominal 90% row-calibrated intervals covered 0.57 (bio-oil; 95% CI 0.52–0.64) and 0.61 (solids) of runs with equal study weights, and 66% and 66% of studies had coverage below 0.80 (Table 4, Fig. 8). Calibrating on out-of-study residuals with equal study weights gave 0.91 (0.87–0.94) and 0.91; the improvement was significant across studies (one-sided Wilcoxon on per-study coverage, $p < 0.001$ for bio-oil and $< 0.001$ for solids). The price is width: 59.1 vs 22.8 wt% for bio-oil and 46.7 vs 20.6 wt% for solids. These widths are not a defect of the method but an honest statement of how little a literature-trained model knows about a study it has not seen. The same pattern held for $\alpha = 0.05$ and 0.20, with 5 instead of 10 outer folds (Supplementary Tables S4–S5), and on the locked test studies (bio-oil RC 0.60 vs SC 0.89; solids RC 0.46 vs SC 0.82), for left-out feedstock families (mean) (bio-oil RC 0.55 vs SC 0.89; solids RC 0.60 vs SC 0.90), for temporally later studies (bio-oil RC 0.42 vs SC 0.85; solids RC 0.56 vs SC 0.90), on the external hydrochar database (solids RC 0.55 vs SC 0.92). An exploratory, post-hoc variant with locally adaptive widths (Supplementary Section S4) did not narrow the intervals (mean width 63.0 vs 59.1 wt% for bio-oil) and worsened the interval score, consistent with an error dominated by an unpredictable study offset.

Whether this holds for other base learners, and what the machine-learning model adds, was tested by repeating the evaluation with seven base learners (E9; Supplementary Table S10). Study-calibrated coverage was 0.90–0.91 for bio-oil and 0.89–0.92 for solids with every base learner, including the training mean. Row-calibrated coverage, by contrast, deteriorated with the capacity of the learner: 0.86 for the mean predictor, 0.80 for ridge regression, 0.65 for *k*-

NN and 0.55–0.59 for the tree ensembles (bio-oil). This is the mechanism behind the failure of row calibration: flexible learners absorb the offsets of studies they have seen, so residuals of randomly held-out runs no longer contain the offset that a new study will carry. Relative to the 'no-ML' reference, the best learner narrowed valid intervals by 15% for bio-oil (53.5 vs 63.0 wt%, random forest) and 23% for solids (46.0 vs 59.4 wt%) — a real but modest gain — and after three local runs the median per-study RMSE was 8.18 vs 9.15 wt% (bio-oil) and 7.85 vs 9.89 wt% (solids).

**Table 4.** Empirical coverage of 90% prediction intervals: row-calibrated (RC) vs study-calibrated (SC).

| Target | Setting | α | Calibration | Coverage (rows) | Coverage (studies) | Width (wt%) | Studies <0.8 coverage |
|---|---|---|---|---|---|---|---|
| Bio-oil | Development CV (unseen studies) | 0.1 | RC | 0.612 | 0.57 | 22.8 | 0.66 |
| Bio-oil | Development CV (unseen studies) | 0.1 | SC | 0.93 | 0.909 | 59.1 | 0.16 |
| Solid | Development CV (unseen studies) | 0.1 | RC | 0.615 | 0.614 | 20.6 | 0.66 |
| Solid | Development CV (unseen studies) | 0.1 | SC | 0.901 | 0.908 | 46.7 | 0.16 |
| Bio-oil | Locked test | 0.1 | RC | 0.565 | 0.603 | 22.7 | 0.61 |
| Bio-oil | Locked test | 0.1 | SC | 0.82 | 0.895 | 54.8 | 0.21 |
| Solid | Locked test | 0.1 | RC | 0.44 | 0.457 | 21.0 | 0.77 |
| Solid | Locked test | 0.1 | SC | 0.864 | 0.821 | 46.7 | 0.23 |
| Bio-oil | Leave-family-out (mean) | 0.1 | RC | 0.569 | 0.554 | 22.5 | 0.62 |
| Bio-oil | Leave-family-out (mean) | 0.1 | SC | 0.888 | 0.894 | 57.6 | 0.16 |
| Bio-oil | Temporal (≤2018 → >2018) | 0.1 | RC | 0.456 | 0.422 | 21.9 | 0.84 |
| Bio-oil | Temporal (≤2018 → >2018) | 0.1 | SC | 0.846 | 0.853 | 64.1 | 0.28 |
| Solid | Leave-family-out (mean) | 0.1 | RC | 0.548 | 0.596 | 20.5 | 0.57 |
| Solid | Leave-family-out (mean) | 0.1 | SC | 0.858 | 0.896 | 46.5 | 0.16 |
| Solid | Temporal | 0.1 | RC | 0.521 | 0.559 | 18.7 | 0.72 |


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| Target | Setting | α | Calibration | Coverage (rows) | Coverage (studies) | Width (wt%) | Studies <0.8 coverage |
|---|---|---|---|---|---|---|---|
| | (≤2018 → >2018) | | | | | | |
| Solid | Temporal (≤2018 → >2018) | 0.1 | SC | 0.877 | 0.897 | 49.0 | 0.14 |
| Solid | External 4TU | 0.1 | RC | 0.534 | 0.549 | 20.7 | 0.7 |
| Solid | External 4TU | 0.1 | SC | 0.934 | 0.918 | 47.3 | 0.16 |

Study calibration works for contrasts too: intervals for within-study contrasts covered 0.90 (bio-oil) and 0.91 (solids) of unseen-study pairs and were 44.9 and 45.2 wt% wide, against 55.8 and 46.6 wt% for levels — narrower for bio-oil, hardly for solids (T4).

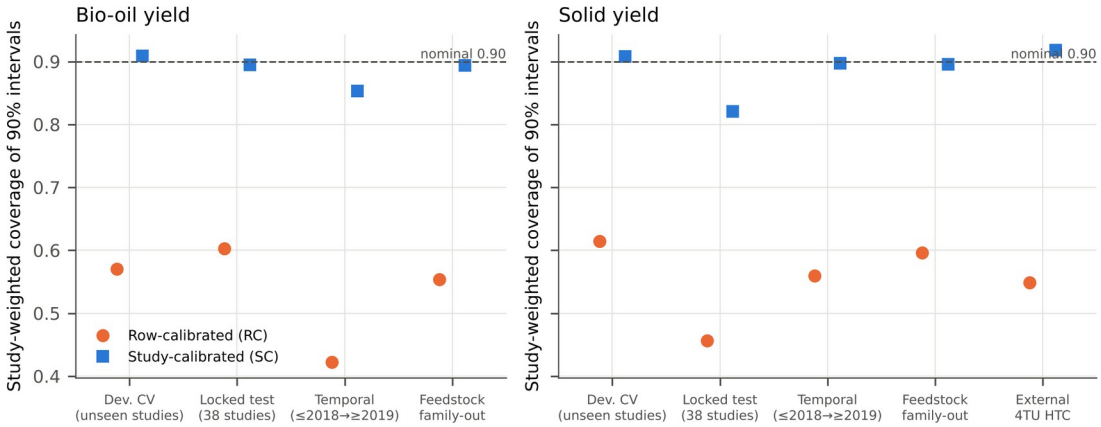



***Fig. 8.*** *Study-weighted empirical coverage of nominal 90% prediction intervals calibrated on random rows (RC) or on studies (SC) across validation settings (base + P features). Dashed line: nominal level.*

## 3.7. A few local runs reduce error and interval width

When *k* runs of the new study are available, its offset can be estimated. For the 80 development studies with at least 11 bio-oil runs, the median per-study RMSE fell from 12.37 wt% (*k* = 0) to 9.22, 8.47 and 7.63 wt% with *k* = 1, 3 and 10 shrunk offsets, and the study-calibrated interval width fell from 59.3 to 37.2 wt% at *k* = 3 while study-weighted coverage stayed at 0.91 (Table 5, Fig. 9). For solids the corresponding RMSE values were 11.05, 8.84,


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7.66 and 7.02 wt%. Shrinkage lowered the per-study RMSE relative to the raw mean offset at k = 1, 2, 3, 5, 10 (Holm-corrected Wilcoxon, bio-oil) and was never significantly worse. The estimated shrinkage constant ($\hat{\lambda}=\hat{\sigma}_w^2/\hat{\tau}^2$, median over folds 0.48 for bio-oil) is small, so the shrunk and raw offsets converge after a few runs; shrinkage matters mainly for *k* = 1–2, which is exactly the regime of a laboratory screening a new feedstock. Estimating $\tau^2$ by restricted maximum likelihood instead of the method of moments gave median per-study RMSE at k = 1 of 9.37 vs 10.06 wt% (bio-oil) and 9.33 vs 8.89 wt% (solids; Supplementary Table S6). Replacing the tuned hyper-parameters of the base learner by library defaults changed its study-grouped RMSE by 0.0–1.0 wt% (Supplementary Table S7), so the conclusions do not hinge on tuning. These results quantify, with valid intervals, the observation of Elfetni et al. (2027) that local calibration runs reduce the error of literature-trained HTL surrogates. Anchoring with offset-invariant models (pre-registered T3) did not improve on the shrunk offset of the absolute model: with one measured run, the median per-study RMSE was 9.60 wt % for STRATA, 10.20 for W and 9.41 for PD (bio-oil, XGBoost; 0 significantly better and 0 significantly worse of 36 Holm-corrected comparisons across learners, targets and *k*).

**Table 5.** k-shot adaptation to a new study (studies with ≥ 11 runs; 20 random support sets per study).

| Target | k | Offset | Median study RMSE | Coverage (studies) | Width (wt%) | Studies | EB vs raw: Δ median; p_Holm |
|---|---|---|---|---|---|---|---|
| Bio-oil | 0 | — | 12.37 | 0.926 | 59.3 | 80 | |
| Bio-oil | 1 | EB shrinkage | 9.22 | 0.908 | 42.6 | 80 | -0.75; <0.001 |
| Bio-oil | 1 | raw mean | 10.23 | 0.907 | 45.9 | 80 | |
| Bio-oil | 2 | EB shrinkage | 8.78 | 0.914 | 39.0 | 80 | -0.29; <0.001 |
| Bio-oil | 2 | raw mean | 9.17 | 0.911 | 39.9 | 80 | |
| Bio-oil | 3 | EB shrinkage | 8.47 | 0.913 | 37.2 | 80 | -0.14; <0.001 |
| Bio-oil | 3 | raw mean | 8.7 | 0.91 | 37.7 | 80 | |
| Bio-oil | 5 | EB shrinkage | 8.19 | 0.918 | 36.3 | 80 | -0.07; <0.001 |


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| Target | k | Offset | Median study RMSE | Coverage (studies) | Width (wt%) | Studies | EB vs raw: Δ median; p_Holm |
|---|---|---|---|---|---|---|---|
| Bio-oil | 5 | raw mean | 8.29 | 0.915 | 36.5 | 80 | |
| Bio-oil | 10 | EB shrinkage | 7.63 | 0.902 | 33.4 | 80 | -0.02; 0.004 |
| Bio-oil | 10 | raw mean | 7.63 | 0.903 | 33.9 | 80 | |
| Solid | 0 | — | 11.05 | 0.898 | 46.6 | 72 | |
| Solid | 1 | EB shrinkage | 8.84 | 0.908 | 39.3 | 72 | -0.92; <0.001 |
| Solid | 1 | raw mean | 9.18 | 0.911 | 45.7 | 72 | |
| Solid | 2 | EB shrinkage | 8.18 | 0.899 | 35.8 | 72 | -0.34; 0.002 |
| Solid | 2 | raw mean | 8.31 | 0.901 | 38.2 | 72 | |
| Solid | 3 | EB shrinkage | 7.66 | 0.899 | 34.6 | 72 | -0.19; 0.003 |
| Solid | 3 | raw mean | 7.87 | 0.9 | 35.9 | 72 | |
| Solid | 5 | EB shrinkage | 7.48 | 0.898 | 33.0 | 72 | -0.09; 0.007 |
| Solid | 5 | raw mean | 7.38 | 0.899 | 33.6 | 72 | |
| Solid | 10 | EB shrinkage | 7.02 | 0.904 | 33.6 | 72 | -0.03; 0.006 |
| Solid | 10 | raw mean | 6.99 | 0.903 | 33.7 | 72 | |

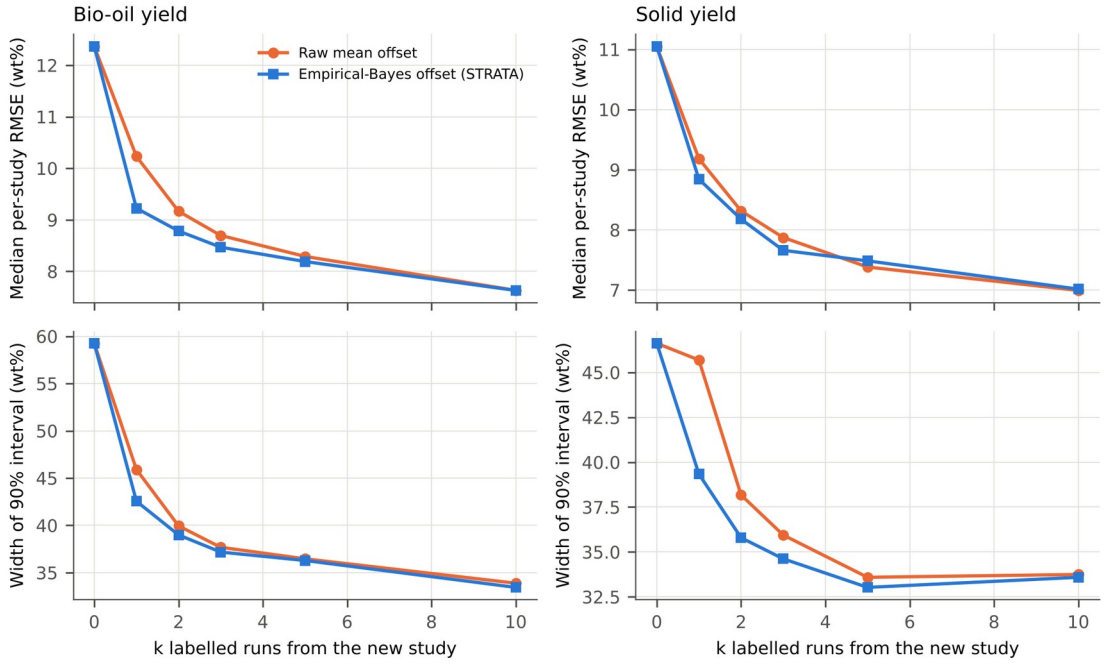



***Fig. 9.*** *k-shot adaptation to a new study (development studies with ≥ 11 runs, base + P features, 20 random support sets per study and k). Top: median per-study RMSE of the remaining runs; bottom: width of the re-calibrated 90% interval. k = 0 is the unadapted model.*


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## 3.8. Generalisation, robustness and cost

The locked test studies, evaluated once after all modelling decisions had been frozen, gave a more pessimistic picture than development CV. For bio-oil yield (556 runs, 28 studies) the STRATA base learner reached RMSE = 18.3 wt% ($R^2$ = -0.95; best learner k-nearest neighbours, 12.9 wt%; mean predictor 13.6 wt%); for solid yield (457 runs, 30 studies) the STRATA base learner reached RMSE = 18.5 wt% ($R^2$ = 0.30; best learner MLP, 14.4 wt%; mean predictor 23.0 wt%). For bio-oil, only 4 of the 11 non-trivial learners had a lower pooled RMSE than the mean predictor, although 6 had a lower median per-study RMSE: pooled errors were dominated by a few studies outside the bulk of the training distribution — for XGBoost, a single solvothermal series of 90 runs contributed 54% of the squared error, and excluding it (post hoc) lowered the RMSE from 18.3 to 13.6 wt% ($R^2$ = 0.00). Study-weighted coverage of row-calibrated 90% intervals was 0.60 (95% CI 0.47–0.73) for bio-oil and 0.46 (95% CI 0.34–0.58) for solids; study-calibrated intervals reached 0.89 (95% CI 0.81–0.97) and 0.82 (95% CI 0.70–0.92) (Table 6, Fig. 8). Study calibration thus removed most, but for solids not all, of the under-coverage; with 28–30 locked studies the intervals are wide, and the solid-yield shortfall reflects a few studies with feedstocks (technical lignin, plastic–biomass blends) that are rare in the training data.

**Table 6.** Single evaluation on the 38 locked test studies (models trained on all development studies, base features).

| Target | Learner | Runs | Studies | RMSE | $R^2$ | Median study RMSE |
|---|---|---|---|---|---|---|
| Bio-oil | k-NN | 556 | 28 | 12.878 | 0.029 | 11.935 |
| Bio-oil | Ridge | 556 | 28 | 12.965 | 0.016 | 11.992 |
| Bio-oil | GPBoost | 556 | 28 | 13.209 | -0.021 | 10.769 |
| Bio-oil | CatBoost | 556 | 28 | 13.389 | -0.049 | 10.778 |
| Bio-oil | Mean | 556 | 28 | 13.634 | -0.088 | 11.123 |
| Bio-oil | SVR | 556 | 28 | 13.913 | -0.133 | 11.882 |
| Bio-oil | Extra trees | 556 | 28 | 14.643 | -0.255 | 11.336 |
| Bio-oil | Random | 556 | 28 | 15.692 | -0.441 | 9.709 |


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| Target | Learner | Runs | Studies | RMSE | $R^2$ | Median study RMSE |
|---|---|---|---|---|---|---|
| | forest | | | | | |
| Bio-oil | XGBoost | 556 | 28 | 18.273 | -0.954 | 9.597 |
| Bio-oil | LightGBM | 556 | 28 | 20.122 | -1.37 | 9.648 |
| Bio-oil | MLP | 556 | 28 | 22.521 | -1.969 | 14.511 |
| Bio-oil | HistGB | 556 | 28 | 22.839 | -2.053 | 9.766 |
| Solid | MLP | 457 | 30 | 14.416 | 0.573 | 10.978 |
| Solid | Extra trees | 457 | 30 | 16.546 | 0.438 | 12.349 |
| Solid | SVR | 457 | 30 | 17.052 | 0.403 | 12.653 |
| Solid | k-NN | 457 | 30 | 18.401 | 0.305 | 11.857 |
| Solid | XGBoost | 457 | 30 | 18.528 | 0.295 | 13.71 |
| Solid | Ridge | 457 | 30 | 18.744 | 0.279 | 14.67 |
| Solid | LightGBM | 457 | 30 | 18.889 | 0.268 | 12.971 |
| Solid | HistGB | 457 | 30 | 19.196 | 0.244 | 13.187 |
| Solid | Random forest | 457 | 30 | 19.316 | 0.234 | 12.56 |
| Solid | CatBoost | 457 | 30 | 19.809 | 0.194 | 13.243 |
| Solid | GPBoost | 457 | 30 | 21.937 | 0.012 | 17.291 |
| Solid | Mean | 457 | 30 | 23.03 | -0.089 | 17.925 |

Predicting studies published after 2018 from earlier studies gave RMSE = 19.4 (bio-oil; higher than the study-grouped CV value of 15.8 wt%) and 16.5 wt% (solids; higher than 14.1 wt%), i.e. models trained on older literature transfer no better to newer studies than to randomly chosen unseen studies. Leaving out one feedstock family at a time gave mean RMSE of 17.2 and 16.3 wt%, with the largest errors for the ‘Other’ category (bio-oil, 32.6 wt %) and ‘Other’ (solids, 29.8 wt%). On the independent 4TU hydrochar database (502 runs after removing 36 potential duplicates; 69 feedstock series), a solid-yield model trained on all development studies reached RMSE = 13.3 wt% ($R^2$ = 0.31), whereas training on the 16 HTC development studies alone gave 16.7 wt% ($R^2$ = -0.08), showing that the small HTC subset of the primary database does not by itself support transfer to other HTC compilations (Table 7).

**Table 7.** Temporal, feedstock-family and external generalisation of the STRATA base learner.

| Target | Features | Scheme | RMSE | $R^2$ | Median study RMSE |
|---|---|---|---|---|---|
| Bio-oil | base | temporal | 19.41 | -0.399 | 14.81 |
| Bio-oil | base+P | temporal | 22.66 | -0.907 | 16.96 |
| Solid | base | temporal | 16.54 | 0.269 | 10.58 |

| Target | Features | Scheme | RMSE | $R^2$ | Median study RMSE |
|---|---|---|---|---|---|
| Solid | base+P | temporal | 16.56 | 0.268 | 10.83 |
| Bio-oil | base | family_out (mean over families) | 17.17 | -0.419 | 12.93 |
| Bio-oil | base+P | family_out (mean over families) | 17.48 | -0.237 | 12.05 |
| Solid | base | family_out (mean over families) | 16.35 | -0.221 | 10.79 |
| Solid | base+P | family_out (mean over families) | 15.5 | -0.03 | 10.28 |
| Solid | base | external_4TU_all_dev_processes | 13.3 | 0.31 | 10.93 |
| Solid | base | external_4TU_HTC_dev_only | 16.66 | -0.082 | 11.59 |

Adding Gaussian noise of 20% of the training standard deviation to continuous test features increased the bio-oil RMSE from 15.7 to 16.6 wt% and the solid RMSE from 13.9 to 15.4 wt %, while SC coverage was 0.90 without and 0.90 with noise (bio-oil) — the error budget for unseen studies is dominated by the study offset, not by input noise. Training on 25% instead of all training studies raised the bio-oil RMSE from 15.7 to 20.3 wt%, and SC coverage remained 0.89, whereas RC coverage stayed at 0.45–0.56 (Supplementary Fig. S1).

All learners trained in at most 9 s on one CPU thread (MLP); the STRATA base learner took 0.2 s and predicted 1,000 runs in 4 ms (Supplementary Table S17, Supplementary Fig. S2). The complete STRATA procedure — two inner five-fold cross-validations, quantiles, variance components and k-shot quantiles for all k — took 2.3–7.0 s on all development studies, against 0.3–0.5 s for fitting the base learner alone (measured, 1 and 4 threads); this overhead is negligible compared with the cost of a single hydrothermal experiment.

Because literature compilations can contain the same experiment reported in several papers, which would leak information across study-grouped folds, the development data were audited (E10): 166 bio-oil runs (7.3%, 16 studies) and 143 solid-yield runs (6.6%) had an identical counterpart in another study. Removing them increased the study-grouped RMSE of the STRATA base learner from 15.8 to 16.6 wt% (bio-oil) and from 14.1 to 14.6 wt% (solids), with between-study shares of 0.65 and 0.52; 10 bio-oil and 18 solid locked-test runs matched


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development runs. Cross-study duplicates therefore make study-grouped estimates slightly optimistic and do not affect the conclusions.

## 3.9. Error analysis

Out-of-study errors were largest for Other feedstocks among families (bio-oil RMSE 33.9 wt %, 6 studies), for the ≤250 °C temperature band (21.1 wt%) and for HTL solvothermal runs (18.5 wt%) (Supplementary Table S9). The five worst-predicted bio-oil studies had per-study RMSE between 35.1 and 61.4 wt% and absolute mean offsets of 26.7–60.9 wt% (processes: HTL 3, HTC 1, HTL solvothermal 1). Across extraction-solvent groups with at least three studies, the bio-oil RMSE ranged from 10.8 (hydrocarbon) to 19.8 wt% (none named) and the mean residual from -9.5 to +9.7 wt% (Supplementary Fig. S3).

## 3.10. What literature-trained models can and cannot do for a new laboratory

Table 8 condenses the evidence. Of the six model capabilities a new laboratory might expect, only two were supported on held-out studies: honest uncertainty, provided intervals are calibrated across studies, and the correction of predictions with a few local runs; the seventh entry, agreement of laboratories on the direction of temperature effects, is a model-free observation. Absolute bio-oil yields of a new study were barely predictable, study offsets were not explained by reported metadata, within-study changes were predicted no better than levels, and the exploratory advantage in choosing conditions did not replicate. The only robust *trend* knowledge was model-free and directional (Fig. 5). Table 9 translates these findings into a minimum validation and reporting checklist for literature-trained bioresource surrogates.


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**Table 8.** What literature-trained hydrothermal models can and cannot do for a new study (development: unseen studies in study-grouped CV; locked: 38 studies held out before modelling).

| Capability | Development studies (unseen) | 38 locked studies | Verdict |
|---|---|---|---|
| Predict the absolute yield of a run in a new study | best $R^2$ 0.24 (bio-oil), 0.54 (solid) | best $R^2$ 0.03 (bio-oil), 0.57 (solid) | Poor (bio-oil); moderate (solid) |
| Explain a study's offset from reported work-up / yield basis | P ablation p = 0.80, 0.54; meta-regression $R^2$ -0.03, -0.05 | 0 of 24 learner–target tests significant | No |
| Predict how yield changes between two runs of a new study (within-study contrast) | contrast $R^2$ 0.14–0.21 (bio-oil), 0.37–0.43 (solid); offset-invariant models better in 1/12 tests | contrast $R^2$ -0.30–0.30 (bio-oil), 0.43–0.56 (solid) | Weak; not better than levels (pre-registered T1–T3 not supported) |
| Direction of the temperature effect below 300 °C (model-free, within studies) | same sign in 81%–95% of studies (bio-oil ↑, solid ↓) | not applicable (descriptive) | Consistent across laboratories |
| Pick the best of a new study's planned runs (bio-oil) | better than T≈325 °C rule (median regret 7.0–8.7 vs 12.8 wt%; exploratory) | not confirmed (p_Holm ≥ 0.62; regret 11.9–12.5 vs 14.6 wt%) | Not established |
| Honest 90% prediction intervals for a new study | row-calibrated 0.57/0.61; study-calibrated 0.91/0.91 | row-calibrated 0.60/0.46; study-calibrated 0.89/0.82 | Yes, if calibrated across studies (wide: ±23–30 wt%) |
| Use 1–3 measured runs of the new study to correct predictions | median study RMSE 12.4 → 8.5 (bio-oil), 11.1 → 7.7 (solid) with 3 runs | median study RMSE 10.9 → 7.8 (bio-oil), 14.2 → 7.4 (solid) | Yes — the most effective use of literature models |

**Table 9.** Minimum validation and reporting checklist for machine-learning models trained on literature-compiled bioresource data.

| Item | Minimum practice | Why (evidence in this study) |
|---|---|---|
| 1. Splitting unit | Hold out whole source studies (or laboratories); report random-split metrics only as a reference | Random splits overstated accuracy about two-fold (Table 2) |
| 2. Locked test | Fix a list of held-out studies before modelling and evaluate it once | Locked results were worse than development CV (Table 6) |
| 3. Duplicates | Audit identical runs reported in several studies before splitting | ~7 % of runs were cross-study duplicates (Section 3.8) |
| 4. Accuracy metrics | Report per-study errors and the mean-predictor baseline next to pooled $R^2$ | Pooled metrics were dominated by a few studies (Section 3.8) |
| 5. Uncertainty | Calibrate prediction intervals across studies, not across random runs | Row-calibrated 90 % intervals covered 0.46–0.61 (Table 4) |
| 6. Trend / optimum claims | Support claims about effects of conditions or optimal conditions with held-out-study contrasts or rankings | Within-study changes and rankings did not transfer (Sections 3.4–3.5) |
| 7. Local anchoring | Recommend 1–3 local runs and report the anchored error | Three runs reduced the median error by 3–4 wt% (Table 5) |
| 8. Metadata | Record product-recovery solvent, yield basis, fraction definitions, mass-balance closure and reactor details in curated databases | Coarse parsed descriptors did not explain offsets (Table 3) |
| 9. Openness | Release code, the held-out study list and per-run predictions | Enables re-analysis of the above |

For practitioners, three recommendations follow. First, literature-trained hydrothermal surrogates should be validated with folds defined by source study (or by laboratory), and random-split metrics should not be used to claim predictive ability for new feedstocks or plants. Second, reported prediction intervals should be calibrated across studies; for the present database this means honest intervals of roughly ±30 wt% for bio-oil yield of a new study, which is wide enough to rule out fine-grained optimisation from literature data alone. Third, a handful of runs on the target feedstock and reactor are worth more than additional model complexity: three local runs reduced the median per-study error by 3.9 wt% (bio-oil) and 3.4 wt% (solids), whereas the median per-study errors of the competitive learners differed by at most 0.6 and 1.6 wt% (Sections 3.1 and 3.7). For data curators, the null result for work-up descriptors suggests recording richer, standardised metadata (product-recovery procedure, quench, mass-balance closure, uncertainty of the yield measurement) so that the sources of between-study variation can eventually be modelled rather than absorbed.

## 3.11. Limitations

The study relies on one curated literature database; its extraction and harmonisation choices, imputed feedstock fractions and residual transcription errors propagate into all results, although they affect random and study-grouped evaluation alike. The protocol descriptors were parsed from free-text fields and are coarse. Study-level conformal validity assumes that a new study is exchangeable with the calibration studies; a study that differs systematically (a new reactor concept, unusual feedstock or catalyst) may be under-covered, as the locked solid-yield studies (coverage 0.82) and the HTC subset of the development studies (coverage 0.83 for solids) show. The external database lacks study identifiers, so feedstock series were used as pseudo-studies. The *k*-shot analysis assumes that the first *k* runs of a study are exchangeable with its remaining runs; in practice laboratories choose runs deliberately.

Physical admissibility across phases was not enforced here; because STRATA only requires out-of-study residuals of a point predictor (Supplementary Table S10), it could be applied to physically admissible surrogates, but this was not tested. Cross-study duplicate runs (Section 3.8) were retained in the main analyses, which makes the reported study-grouped accuracy slightly optimistic. Three analyses (E9–E11), the adaptive-interval variant and the locked-test sensitivity analysis were added after the main results were known. Finally, no new experiments were performed: the conclusions concern the transferability of literature-trained models, not the chemistry of hydrothermal conversion itself.

# 4. Conclusions

Machine-learning models trained on published hydrothermal-conversion data predict runs of studies they have seen far better than runs of a new study: random splitting overstated accuracy about two-fold, and on locked studies no learner predicted bio-oil yields substantially better than the training mean. Half to two-thirds of the out-of-study error variance was a study-level offset that recorded work-up and yield-basis descriptors did not explain. Laboratories agree on the direction of the temperature response, but models did not predict within-study changes better than absolute yields, and an exploratory advantage in choosing conditions did not replicate on held-out studies. What did hold was honest uncertainty — conformal intervals calibrated across studies raised coverage from 0.46–0.61 to 0.82–0.91 — and the correction of predictions with a few local runs. Study-grouped validation with a locked test set, study-calibrated intervals and local anchoring runs are recommended as minimum practice (Table 9), and claims about trends or optimal conditions should be supported by held-out-study evidence.


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## CRediT authorship contribution statement

[Author 1]: Conceptualization, Methodology, Software, Formal analysis, Writing – original draft. [Author 2]: Validation, Writing – review & editing. [Author 3]: Supervision, Writing – review & editing. *(To be completed by the authors.)*

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper. *(To be confirmed by the authors.)*

## Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the authors used an AI assistant (Claude, Anthropic) to search and verify literature metadata, write analysis code, run the computational experiments and draft the manuscript text. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article. *(The authors must verify, adapt and confirm this statement before submission.)*

## Data availability

All data analysed are publicly available under CC BY 4.0 (Elfetni, 2026; Abdeldayem et al., 2025). The complete code, the locked test-study list, configuration files, raw result files and the scripts that generate every table and figure are available at [repository URL to be inserted; archived DOI to be minted at acceptance].


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## Appendix A. Supplementary data

Supplementary material (Sections S1–S9, Tables S1–S17, Figs. S1–S4: literature screening, protocol and addenda, additional benchmark and sensitivity results, exploratory adaptive intervals, error analysis by subgroup, STRATA with different base learners, trend and ranking analyses, reproducibility) accompanies this article.

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